Syllabus Details

Syllabus ID:	14433
Syllabus Name:	**Microcontroller Programming_Lập trình vi điều khiển**
Course Name English:	**Microcontroller Programming**
Subject Code:	**MIP201**
Learning-Teaching Method:	In-class lecture, Active Learning, Social Constructivism, Project-Based Learning,
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours + 1h final exam + 104h self-study
Pre-Requisite:	PRF192
Description:	Hands-on Embedded Systems Development: Learn to implement real-world embedded system projects on microcontrollers (using both simulation environments and physical hardware) with the STM32Cube HAL library and STM32CubeIDE. Peripheral Mastery & ARM Architecture: Master the core architecture of the ARM Cortex-M processor. Develop and refine C/C++ source code to build diverse applications (signal generators, temperature monitoring systems, audio players, and user interfaces) by leveraging Digital/Analog I/O, Interrupts, Timers, and Pulse-Width Modulation (PWM). Gain hands-on proficiency in analyzing and deploying serial communication protocols—SPI, UART, and I2C—while evaluating their advantages and limitations in embedded environments. Real-Time Operating Systems (RTOS): Explore the core characteristics of RTOS and multitasking mechanisms on microcontrollers. Students will directly apply Threads, Mutexes, Semaphores, and Interrupts to design a complete multitasking system, effectively resolving resource contention issues. Practical Engineering Skills: Prototype embedded systems using the ST Nucleo-F401RE (ST Nucleo-F411RE) development board and cost-effective components. Enhance hardware-level C/C++ programming mindsets by immediately applying theory to practice. Master the end-to-end hardware development workflow alongside professional debugging and troubleshooting techniques.
StudentTasks:	"- Students must attend at least 80% of contact sessions in order to be accepted to the final examination. - Student is responsible to do all exercises and practical given by instructor in class or at home and submit on time. Do quizzes during class. - Students must regularly check the Learning Management System (http://flm.fpt.edu.vn and ehttps://fu-edunext.fpt.edu.vn/home) for official announcements, schedule updates, and course materials."
Tools:	Specific Tools: 1. Hardware & Training Kits - Development Board: ST Nucleo-F401RE (powered by the ARM Cortex-M4 core). - Programmer & Debugger: On-board ST-Link/V2 debugger, supporting the SWD (Serial Wire Debug) protocol. 2. Software Development Environment - Primary IDE: STM32CubeIDE (fully replaces Keil Studio Cloud and Mbed Studio). This is the industry-standard IDE for C/C++ development on ST microcontrollers. - Simulation Software: Wokwi (a free, web-based open-source simulator supporting STM32 boards and various peripherals) or other debugging/simulation tools recommended by the instructor.
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	968/QĐ-ĐHFPT dated 08/07/2026
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	8/7/2026

5 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	A Beginner's Guide to Designing Embedded System Applications on Arm Cortex-M Microcontrollers	Ariel Lutenberg, Pablo Gomez, Eric Pernia	ARM Education Media	2022		9781911531418	True	False	True	https://www.arm.com/res
2	Getting started with STM32: STM32 step-by-step	STMicroelectronics	STMicroelectronics	2026		N/A	False	False	True	https://wiki.st.com/stm32
3	Getting started with STM32 system peripherals	STMicroelectronics	STMicroelectronics	2026		N/A	False	False	True	https://wiki.st.com/stm32
4	STM32 Programming Tutorials & Projects	Khaled Magdy	DeepBlueMbedded	N/A		N/A	False	False	True	https://deepbluembedde
5	STM32 Tutorials	Arun Rawat	ControllersTech	N/A		N/A	False	False	True	https://controllerstech.co

10 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Understand the fundamentals of an embedded system and discover the ARM architecture and microcontrollers.
2	CLO2	Understand the characteristics of microcontrollers and embedded systems, including their components, benefits, and constraints.
3	CLO3	Understand the industry-standard ARM-based microcontroller architecture, its features, and programming models.
4	CLO4	Develop working embedded prototypes using the STM32 HAL library, FreeRTOS, the ST Nucleo-F401RE board (or ST Nucleo-F411RE board), and low-cost peripherals.
5	CLO5	Apply knowledge of digital and analog I/O programming on STM32 microcontrollers.
6	CLO6	Implement interrupts to control peripherals using the STM32 HAL library.
7	CLO7	Develop applications incorporating timers and Pulse Width Modulation (PWM) using the STM32 HAL library.
8	CLO8	Evaluate serial communication protocols used in embedded systems (UART, SPI, and I2C), including their benefits and limitations.
9	CLO9	Understand Real-Time Operating Systems (FreeRTOS) and how they facilitate multitasking in embedded systems.
10	CLO10	Implement real-world embedded system projects on the STM32 Nucleo board using FreeRTOS, with proper task synchronization (Mutex/Semaphore).

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Course Overview	Offline	CLO1, CLO2	IT	Books, Slide Chapter 1	MCP201	Read Slide Chapter 1	
2	Chapter 1: Introduction to Embedded Systems 1.1. CPUs vs. MCUs vs. Embedded Systems	Offline	CLO1, CLO2	T	Books, Slide Chapter 1		Read Slide Chapter 1	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
3	1.2. Examples of Embedded Systems 1.3. Options for Building Embedded Systems	Offline	CLO1, CLO2	IT	Books, Slide Chapter 1		Read Slide Chapter 1	
4	1.4. Features of Embedded Systems 1.5. Building Embedded Systems	Offline	CLO1, CLO2	T	Books, Slide Chapter 1		Read Slide Chapter 1	
5	Chapter 2: ARM Cortex-M4 Processor Architecture 2.1. What is Arm architecture? 2.2. Arm processor families	Offline	CLO1, CLO3	IT	Books, Slide Chapter 2		Read Slide Chapter 2	
6	2.3. Arm Cortex-M series 2.4. Cortex-M4 processor	Offline	CLO3	T	Books, Slide Chapter 2		Read Slide Chapter 2	
7	2.5. Arm processors vs. Arm architectures 2.6. Processor overview	Offline	CLO3	T	Books, Slide Chapter 2		Read Slide Chapter 2	
8	2.7. Block diagram 2.8. Registers	Offline	CLO3	T	Books, Slide Chapter 2		Read Slide Chapter 2	
9	2.9. ALU, Pipeline and Instruction Execution	Offline	CLO3	T	Books, Slide Chapter 2		Read Slide Chapter 2	
10	Chapter 3: Memory Mapping & Address Space 3.1. Memory Map of STM32	Offline	CLO3	IT	Books, Slide Chapter 3		Read Slide Chapter 3	
11	3.2. Instruction Cycles and Bus Architecture	Offline	CLO3	T	Books, Slide Chapter 3		Read Slide Chapter 3	
12	Chapter 4: STM32CubeIDE & HAL Library 4.1. Introduction to STM32CubeIDE	Offline	CLO4	IT	Books, Slide Chapter 4		Read Slide Chapter 4	
13	4.2. STM32CubeMX Configuration & Code Generation	Offline	CLO4	T	Books, Slide Chapter 4		Read Slide Chapter 4	
14	4.3. CMSIS-CORE & HAL Driver Architecture	Offline	CLO4	T	Books, Slide Chapter 4		Read Slide Chapter 4	
15	Chapter 5: Digital I/O (GPIO) 5.1. Bare-metal Register Manipulation	Offline	CLO5	IT	Books, Slide Chapter 5		Read Slide Chapter 5	
16	5.2. HAL API for GPIO	Offline	CLO5	T	Books, Slide Chapter 5		Read Slide Chapter 5	
17	Lab 1	Offline	CLO5	TU	Lab 1 tutorials		Completing the Assignment/Lab	
18	Lab 1	Offline	CLO5	TU	Lab 1 tutorials		Completing the Assignment/Lab	
19	Lab 1 Review & Q&A	Offline	CLO5	TU	Lab 1 tutorials		Submit Assignment/Lab 1	
20	Chapter 6: Interrupts & NVIC 6.1. External Interrupts (EXTI)	Offline	CLO6	IT	Books, Slide Chapter 6		Read Slide Chapter 6	
21	6.2. NVIC Configuration and Priority	Offline	CLO6	T	Books, Slide Chapter 6		Read Slide Chapter 6	
22	6.3. Interrupt Service Routine (ISR)	Offline	CLO6	T	Books, Slide Chapter 6		Read Slide Chapter 6	
23	6.4. Race Condition Concepts	Offline	CLO6	T	Books, Slide Chapter 6		Read Slide Chapter 6	
24	Lab 2	Offline	CLO6	TU	Lab 2 tutorials		Completing the Assignment/Lab	
25	Lab 2	Offline	CLO6	TU	Lab 2 tutorials		Completing the Assignment/Lab	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
26	Lab 2 Review & Q&A	Offline	CLO6	TU	Lab 2 tutorials		Submit Assignment/Lab 2	
27	Chapter 7: Timers & PWM 7.1. Hardware Timer Configuration	Offline	CLO7	IT	Books, Slide Chapter 7		Read Slide Chapter 7	
28	7.2. Pulse-Width Modulation (PWM) Theory	Offline	CLO7	T	Books, Slide Chapter 7		Read Slide Chapter 7	
29	Project Kick-off & Requirements	Offline	CLO4, CLO10	TU	Project tutorials		Register Topic	
30	Project Topic Registration & Team Formation	Offline	CLO10	TU	Project tutorials		Submit Design	
31	Project Design – Task Decomposition	Offline	CLO10	TU	Project tutorials		Implement Project	
32	Lab 3	Offline	CLO7	TU	Lab 3 tutorials		Completing the Assignment/Lab	
33	Lab 3	Offline	CLO7	TU	Lab 3 tutorials		Completing the Assignment/Lab	
34	Lab 3 Review & Q&A	Offline	CLO7	TU	Lab 3 tutorials		Submit Assignment/Lab 3	
35	Chapter 8: Analog I/O (ADC) 8.1. Quantization and Vref	Offline	CLO5	T	Books, Slide Chapter 8		Read Slide Chapter 8	
36	8.2. ADC Multi-channel & Scan Mode	Offline	CLO5	T	Books, Slide Chapter 8		Read Slide Chapter 8	
37	8.3. DMA for ADC Data Transfer	Offline	CLO5	T	Books, Slide Chapter 8		Read Slide Chapter 8	
38	8.4. Signal Filtering (Moving Average)	Offline	CLO5	T	Books, Slide Chapter 8		Read Slide Chapter 8	
39	Lab 4	Offline	CLO5	TU	Lab 4 tutorials		Completing the Assignment/Lab	
40	Lab 4	Offline	CLO5	TU	Lab 4 tutorials		Completing the Assignment/Lab	
41	Lab 4 Review & Q&A	Offline	CLO5	TU	Lab 4 tutorials		Submit Assignment/Lab 4	
42	Chapter 9: Serial Communication 9.1. UART Protocol	Offline	CLO8	IT	Books, Slide Chapter 9		Read Slide Chapter 9	
43	9.2. SPI Protocol	Offline	CLO8	T	Books, Slide Chapter 9		Read Slide Chapter 9	
44	9.3. I2C Protocol	Offline	CLO8	T	Books, Slide Chapter 9		Read Slide Chapter 9	
45	Lab 5	Offline	CLO8	TU	Lab 5 tutorials		Completing the Assignment/Lab	
46	Lab 5	Offline	CLO8	TU	Lab 5 tutorials		Completing the Assignment/Lab	
47	Lab 5 Review & Q&A	Offline	CLO8	TU	Lab 5 tutorials		Submit Assignment/Lab 5	
48	Chapter 10: FreeRTOS 10.1. Introduction to RTOS vs while(1)	Offline	CLO9	IT	Books, Slide Chapter 10		Read Slide Chapter 10	
49	10.2. Task Creation and Scheduling	Offline	CLO9	T	Books, Slide Chapter 10		Read Slide Chapter 10	
50	10.3. Mutex – Protecting Shared Resources	Offline	CLO9	T	Books, Slide Chapter 10		Read Slide Chapter 10	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
51	10.4. Semaphore and Queue	Offline	CLO9	T	Books, Slide Chapter 10		Read Slide Chapter 10	
52	10.5. Deadlock Concepts and Avoidance	Offline	CLO9	T	Books, Slide Chapter 10		Read Slide Chapter 10	
53	Lab 6	Offline	CLO9	U	Lab 6 tutorials		Completing the Assignment/Lab	
54	Lab 6	Offline	CLO9	U	Lab 6 tutorials		Completing the Assignment/Lab	
55	Lab 6 Review & Q&A	Offline	CLO9	U	Lab 6 tutorials		Submit Assignment/Lab 6	
56	Project Demonstration + Q&A + Course Wrap-up	Offline	CLO4 - CLO10	U	Project tutorials		Presentation	
57	Project Demonstration + Q&A + Course Wrap-up	Offline	CLO4 - CLO10	U	Project tutorials		Presentation	
58	Project Demonstration + Q&A + Course Wrap-up	Offline	CLO4 - CLO10	U	Project tutorials		Presentation	
59	Project Demonstration + Q&A + Course Wrap-up	Offline	CLO4 - CLO10	U	Project tutorials		Presentation	
60	Project Demonstration + Q&A + Course Wrap-up	Offline	CLO4 - CLO10	U	Project tutorials		Presentation	

0 Constructive question(s)

3 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Assignment/Lab	on-going	2	30.0%	>0	At home and in schedule	All CLOs	Mini-project	By instructor	Assignment/Lab1: Covering knowledge from Chapter 1 to Chapter 4 Assignment/Lab2: Covering knowledge from Chapter 4 to Chapter 5 Assignment/Lab3: Covering knowledge from Chapter 5 to Chapter 7 Assignment/Lab4: Covering knowledge from Chapter 6 to Chapter 8 Assignment/Lab5: Covering knowledge from Chapter 5 to Chapter 9 Assignment/Lab6: Covering knowledge from Chapter 8 to Chapter 10	Supervised by instructor	Pre-Lab (Mandatory): - Setup project in STM32CubeIDE & configure .ioc file. - Write HAL logic in C/C++ and ensure zero compilation errors. In-Lab: - Connect STM32 Nucleo-F401RE(or STM32 Nucleo-F411RE board) to hardware & flash code. - Test physical system behavior and clear the instructor Q&A session. - The instructor will evaluate and grade the student's lab work. Each student must have at least two lab grades.
2	Project	on-going	1	30.0%	4	At home and in schedule	All CLOs	Presentation	By instructor	All chapters; knowledge and skills (Hardware Demo)	Supervised by instructor	
3	Final exam	Final exam	1	40.0%	4	60'	All CLOs	Multiple choice	50	All chapters; knowledge and skills	By Exam board	The exam questions must be updated